ML Assignment 2

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Section A (Theoretical)

Ques 1:

Q.1) $P(\text{Dividend} = \text{Yes}) = \frac{80}{100}$ $\quad$ $P(\text{Dividend} = \text{No}) = \frac{20}{100}$

$$P(\text{Last Year's profit} \mid \text{dividend} = \text{Yes}) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(u-\mu)^2}{2\sigma^2}}$$

→ Gaussian Distribution $\quad \mu = 10$, $\sigma^2 = 36$, $\mu' = 0$

Acc. to Bayes'

$$P(\text{Dividend} = \text{Yes} \mid \text{Last Year's Profit} = 4\%) = \frac{P(\text{Last Year's Profit} = 4\% \mid \text{Dividend} = \text{Yes}) \cdot P(\text{Yes})}{P(\text{Last Year's Profit} = 4\%)} \quad -①$$

D → Dividend

LYP → Last Year's Profit

$$P(\text{Yes}) = \frac{80}{100} \quad -②$$

$$P(LYP = 4 \mid D = \text{Yes}) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(u-\mu)^2}{2\sigma^2}}$$
$$= \frac{1}{\sqrt{2\pi \times 36}} \times e^{-\frac{(4-10)^2}{2\times 36}}$$
$$= \frac{1}{15.04} \times e^{-1/2}$$
$$= \frac{0.606}{15.04} \approx 0.0402 \quad -③$$

$$P(LYP = 4 \mid D = \text{No}) = \frac{1}{\sqrt{2\pi \times 36}} \times e^{-\frac{(4-0)^2}{2\times 36}}$$
$$= \frac{0.8007}{15.04}$$
$$\approx 0.0532$$

$$P(LYP = 4\%) = P(LYP = 4\% \mid D = \text{Yes}) \times P(D = \text{Yes}) + P(LYP = 4\% \mid D = \text{No}) \times P(D = \text{No})$$
$$= (0.0402 \times 0.8) + (0.0532 \times 0.2)$$
$$= 0.03216 + 0.01064$$
$$\approx 0.0428 \quad -④$$

Using ②, ③, ④ in ①

$$P(D = \text{Yes} \mid LYP = 4\%) = \frac{0.0402 \times 0.8}{0.0428}$$
$$\approx 0.7514$$

Probability that a company will issue dividends given 4% profit increase ≈ 0.7514
$\approx 75.14\%$

Ques 2:

Q.2)

Class Time	Had Proper Sleep	Weather	Attended ML Class
Morning	Yes	Cool	Yes
Morning	No	Rainy	No
Morning	No	Cool	Yes
Morning	Yes	Hot	Yes
Noon	Yes	Cool	Yes
Noon	No	Hot	No
Noon	No	Cool	No
Noon	Yes	Hot	Yes
Afternoon	Yes	Cool	Yes
Afternoon	No	Rainy	No
Afternoon	No	Hot	No
Afternoon	Yes	Hot	Yes

$P(Yes) = \frac{7}{12}$ $\quad P(No) = \frac{5}{12}$

$H(S) = -\frac{7}{12} \times \log_2 \frac{7}{12} - \frac{5}{12} \times \log_2 \frac{5}{12}$

$= 0.979$

① Class Time:

Morning	Noon	Afternoon
Y - 3	Y - 2	Y - 2
N - 1	N - 2	N - 2

$H(Morning) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$

$= 0.81125$

$H(Noon) = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4}$

$= 1$

$H(Afternoon) = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4}$

$= 1$

$IG(class\ Time) = 0.979 - \frac{4}{12} \times 0.81125 - \frac{4}{12} \times 1 - \frac{4}{12} \times 1$

≈ 0.042

② Had Proper Sleep:

Yes	No
Y - 6	Y - 1
N 0	N - 5

$H(Yes) = -\frac{6}{6} \log_2 \frac{6}{6} - (0)$

$= 0$

$H(No) = -\frac{1}{6} \log_2 \frac{1}{6} - \frac{5}{6} \log_2 \frac{5}{6}$

$= \cancel{\quad} 0.649$

$IG(Had\ Proper\ Sleep) = 0.979 - \frac{6}{12} \times 0 - \frac{6}{12} \times \cancel{\quad} 0.649$

$\approx \cancel{\quad} 0.654$

③ Weather:

Cool	Rainy	Hot
Y - 4	Y - 0	Y - 3
N - 1	N - 2	N - 2

$H(cool) = -\frac{4}{5} \log_2 \frac{4}{5} - \frac{1}{5} \log_2 \frac{1}{5}$

$= 0.722$

$H(\text{Rainy}) = 0 - 0$
$= 0$

$H(\text{Hot}) = -\frac{3}{5}\log_2\frac{3}{5} - \frac{2}{5}\log_2\frac{2}{5}$

$= 0.971$

$IG(\text{Weather}) = 0.979 - \frac{5}{12} \times 0.722 - 0 - \frac{5}{12} \times 0.971$

≈ 0.274

→ Had Proper Sleep is the best split as highest info gain and if Had Proper Sleep = Yes then Attended ML Class = Yes

Class Time	Had Proper Sleep	Weather	Attended ML Class
Morning	No	Rainy	No
Morning	No	Cool	Yes
Noon	No	Hot	No
Noon	No	Cool	No
Afternoon	No	Rainy	No
Afternoon	No	Hot	No

$P(\text{Yes}) = \frac{1}{6}$ $\quad P(\text{No}) = \frac{5}{6}$

$H(S) = -\frac{1}{6}\log_2\frac{1}{6} - \frac{5}{6}\log_2\frac{5}{6}$

$= 0.649$

① Class Time:

	Morning	Noon	Afternoon
	Y - 1	Y - 0	Y - 0
	N - 1	N - 2	N - 2

$H(\text{Morning}) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}$

$= 1$

$H(\text{Noon}) = 0 - 0$
$= 0$

$H(\text{Afternoon}) = 0 - 0$
$= 0$

$IG(\text{Class Time}) = 0.649 - \frac{2}{6} \times 1 - 0 - 0$

≈ 0.315

② Weather:

	Rainy	Hot	Cool
	Y - 0	Y - 0	Y - 1
	N - 2	N - 2	N - 1

$H(\text{Rainy}) = 0 - 0$
$= 0$

$H(\text{Hot}) = 0 - 0$
$= 0$

$H(\text{Cool}) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2}$
$= 1$

$IG\ (\text{Weather}) = 0.649 - 0 - 0 - \frac{2}{6} \times 1$

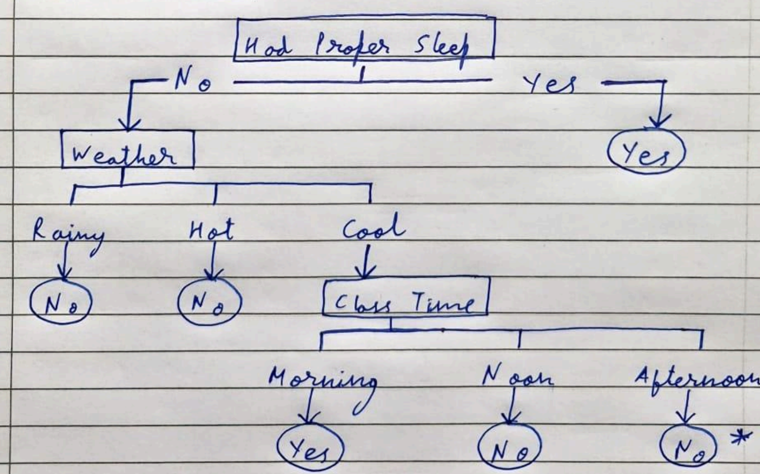
≈ 0.315

→ Can split on either Weather / Class Time as both have same IG

splitting Weather : Rainy → Didn't Attend
: Hot → Didn't Attend

splitting Class Time when Weather is Cool :
Morning → Attended
Noon → Didn't Attend

<u>Decision Tree</u> :



* Assumed No coz no data

Ques 3:

Q3) To Prove : ① No. of updates is at most $\frac{8}{\gamma^2}$

② Derive mistake bound when threshold is $(1-\epsilon)\gamma$

$w_i \rightarrow$ weights after i updates
$w_0 \rightarrow$ optimal weights

Initially, $\|w_1\| = 0$

after n updates

$\Rightarrow \|w_i\| \geq \frac{2}{\gamma}$ — ①

$$w_n \times w_0 \geq n\gamma$$

For a normal perceptron

$\Rightarrow \|w_{i+1}\|^2 \leq \|w_i\|^2 + 1$ — ② $\{\gamma \leq 1\}$

Updated even when margin is less than $\frac{\gamma}{2}$ acc. to Ques

$$\Rightarrow \|w_{i+1}\| \leq \|w_i\| + \frac{1}{2\|w_i\|} + \frac{\gamma}{2}$$

$$\Rightarrow \|w_{i+1}\| \leq \|w_i\| + \frac{\gamma}{4} + \frac{\gamma}{2}$$

$$\Rightarrow \|w_{i+1}\| \leq \|w_i\| + \frac{3\gamma}{4}$$

Given, there have been n updates

$$\|w_{n+1}\| \leq \frac{2}{\gamma} + \frac{3n\gamma}{4} \quad \text{— ③}$$

Using Cauchy Schwarz Inequality

$$w_n \cdot w_0 \leq \|w_n\|$$

$$\Rightarrow n\gamma \leq \frac{2}{\gamma} + \frac{3n\gamma}{4}$$

$$\Rightarrow n - \frac{3n}{4} \leq \frac{2}{\gamma^2}$$

$$\Rightarrow \frac{n}{4} \leq \frac{2}{\gamma^2}$$

$$\Rightarrow n \leq \frac{8}{\gamma^2}$$

① Proved

Given threshold is $(1-\epsilon)\gamma$

$$\Rightarrow w_{i+1} w_0 \geq w_i w_0 + (1-\epsilon)\gamma$$

Also that $\|w_n\|^2 \leq n$ by induction

$\{(1-\epsilon)\gamma \leq 1\}$

Using Cauchy Schwarz

$$n(1-\epsilon)\gamma \leq \|w_n\|$$

$$\Rightarrow n(1-\epsilon)\gamma \leq \sqrt{n}$$

$$\Rightarrow n^2(1-\epsilon)^2\gamma^2 \leq n$$

$$\Rightarrow n \leq \frac{1}{(1-\epsilon)^2\gamma^2}$$

$\frac{1}{(1-\epsilon)^2\gamma^2}$ is the corresponding mistake bound when $\frac{\gamma}{2}$ is replaced by $(1-\epsilon)\gamma$

Ques 4:

Q.4) a) $P(\text{spam}) = 0.5$

$P(\text{not spam}) = 0.5$

$P(\text{buy}=1 \mid \text{spam}) = 1$

$P(\text{buy}=0 \mid \text{spam}) = 0$

$P(\text{buy}=1 \mid \text{not spam}) = 0.5$

$P(\text{buy}=0 \mid \text{not spam}) = 0.5$

$P(\text{cheap}=1 \mid \text{spam}) = 0.5$

$P(\text{cheap}=0 \mid \text{spam}) = 0.5$

$P(\text{cheap}=1 \mid \text{not spam}) = 0.5$

$P(\text{cheap}=0 \mid \text{not spam}) = 0.5$

b) $P(\text{spam} \mid \text{cheap but not buy})$

$$= \frac{P(\text{buy}=0 \mid \text{spam}) \times P(\text{cheap}=1 \mid \text{spam}) \times P(\text{spam})}{P(\text{buy}=0, \text{cheap}=1)}$$

$= 0$

$P(\text{not spam} \mid \text{cheap but not buy})$

$$= \frac{P(\text{buy}=0 \mid \text{not spam}) \times P(\text{cheap}=1 \mid \text{not spam}) \times P(\text{not spam})}{P(\text{buy}=0, \text{cheap}=1)}$$

$$= \frac{0.125}{0.125}$$

$= 1$

c) The problem with zero probabilities is that if a feature never occurs, it makes the probability of an entire class occuring 0.

To prevent this we do:

1. Laplace Smoothing: Add a constt. 1 to numerator and denominator so they aren't zero.
2. m-estimate: Here, we add a prior probability multiplied by a parameter m so they aren't 0.

Section C (Algorithm Implementation using Packages)

Ques 1 (A)

For the given dataset, there are 15 distinct classes with exactly 840 images per class, which means it is well-balanced. The dataset contains no missing or null values so all images are available for analysis without the need for preprocessing.

```
Number of images per class:
label
sitting               840
using_laptop          840
hugging               840
sleeping              840
drinking              840
clapping              840
dancing               840
cycling               840
calling               840
laughing              840
eating                840
fighting              840
listening_to_music    840
running               840
texting               840
Name: count, dtype: int64
```

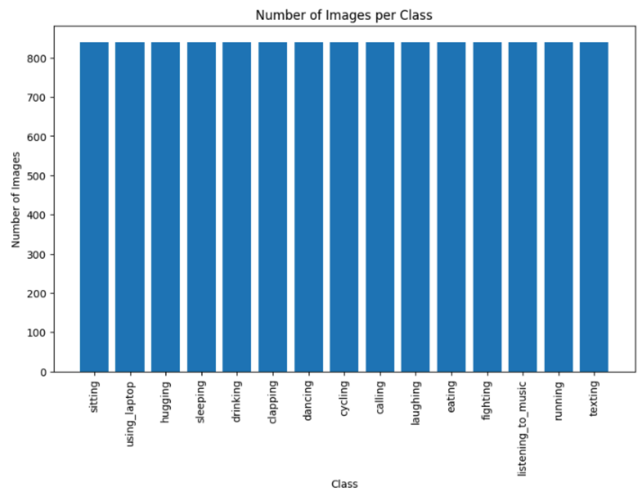
The scatter plot and images below show how the height and width of the images in the dataset are related and also the statistics, indicating that as the height of an image increases, its width generally decreases, with most images falling within specific size ranges. It is important to note that the sizes of the images vary and they will need to be resized later to ensure the feature array of all images is the same.


```
Min Width:   84
Max Width:   478
Mean Width:   260.38103174603174
Std Width:   39.91769674243761
```

```
Min Height:   84
Max Height:   318
Mean Height:   196.57357142857143
Std Height:   35.28000220378205
```

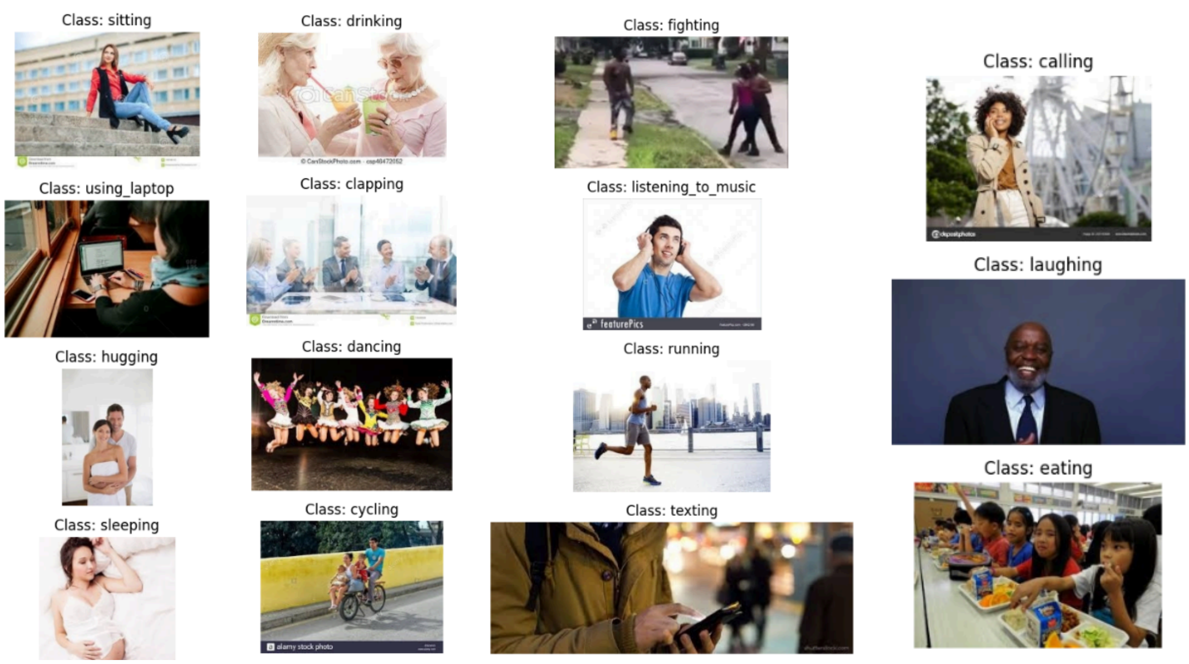
Ques 1 (B)

Distribution of Classes:



The distribution of classes is uniform, with each of the 15 classes containing exactly 840 images making it a balanced dataset.

Sample Images from Each Class:



The data we have is labeled. A sample image from each class can be seen above.

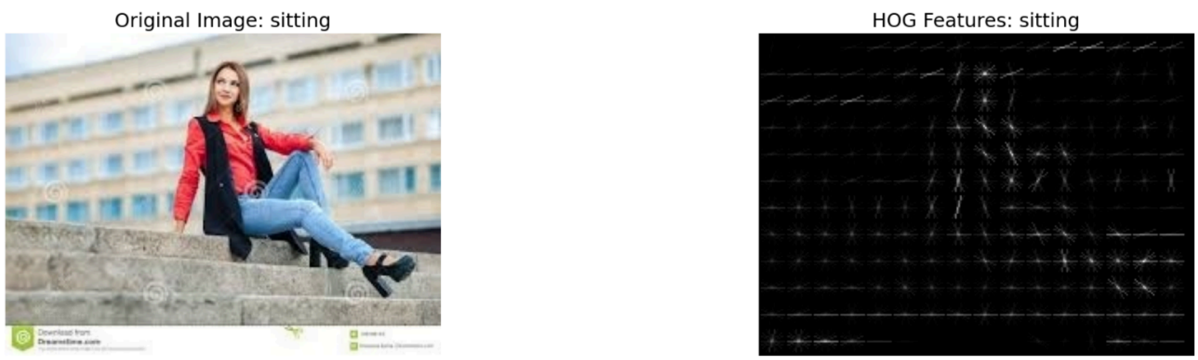
Ques 1 (C)

After investigating the class distribution, I found that the dataset is balanced with 840 images per class. There is no need for techniques like data augmentation or resampling. However, if imbalances were present, data resampling would've helped balance the dataset.

Ques 2

I used three feature extraction methods: Histogram of Oriented Gradients (HOG), Color Histograms, and Local Binary Patterns (LBP).

- **HOG** captures the shape and structure of objects by computing the gradient orientation distribution which helps in detecting edges and contours in the image. Example of a HOG image representing the structure and shape of objects in the image:



- **Color histograms** represent the distribution of colors in the image which provides information about the color composition and helps distinguish between images with different color schemes.
- **LBP** captures the texture by analyzing the local contrast and patterns in small regions of the image making it useful for texture based image classification.

However, I had to keep on reducing the number of features extracted from each method, as the original feature sets would have required an immense amount of computational power and time which was not feasible at this point of time.

The feature extraction process initially yielded around 15,000 features which I successfully reduced to 2,798 by limiting the number of features calculated from each method. The code limits the features by adjusting parameters like orientations and cell sizes for HOG, reducing the number of bins for color histograms, and modifying the points and radius for LBP. Below are the numbers of features extracted from each method used:

```
Processing Image No. 2499
HOG Feature Count: 1764
Color Histogram Feature Count: 1024
LBP Feature Count: 10

Processing Image No. 2500
HOG Feature Count: 1764
Color Histogram Feature Count: 1024
LBP Feature Count: 10

Processing Image No. 2501
HOG Feature Count: 1764
Color Histogram Feature Count: 1024
LBP Feature Count: 10
```

Ques 3 (A)

I used Naive Bayes, Decision Trees, Random Forest, Perceptron, and XGBoost to train and test the models on the prepared dataset with an 80:20 train-test split. Below are the sizes of the X, Y, X_train, and X_test sets:

```
Size of X (features): (12600, 2798)
Size of y (labels): 12600
Size of X_train (scaled): (10080, 2798)
Size of X_test (scaled): (2520, 2798)
```

Ques 3 (B)

```
1. Naive Bayes
Naive Bayes Accuracy: 0.16785714285714284
2. Decision Tree
Decision Tree Accuracy: 0.18253968253968253
3. Perceptron
Perceptron Accuracy: 0.2563492063492063
4. Random Forest
Random Forest Accuracy: 0.32976190476190476
```

```
/usr/local/lib/python3.10/dist-packages/xgboost/core.py:158: UserWarning: [12:06:03] WARNING: /workspace/src/learner.cc:740:
Parameters: { "use_label_encoder" } are not used.

  warnings.warn(smsg, UserWarning)
XGBoost Accuracy: 0.4087301587301587
```

The accuracies for each model are as follows:

- Naive Bayes: 16.786%
- Decision Tree: 18.254%
- Perceptron: 25.635%
- Random Forest: 32.976%
- XGBoost: 40.873%

Given that XGBoost and Random Forest had the best accuracies, I decided to use an ensemble of the two with soft voting which combines the predicted probabilities by averaging them for each class and then selecting the class with the highest average probability as the final prediction. This approach resulted in the best accuracy of **41.587%** as seen below:

```
/usr/local/lib/python3.10/dist-packages/xgboost/core.py:158: UserWarning: [12:33:53] WARNING: /workspace/src/learner.cc:740:
Parameters: { "use_label_encoder" } are not used.

  warnings.warn(smsg, UserWarning)
Ensemble Model Accuracy: 0.4158730158730159
```

Ques 4

I saved and loaded the top 3 models using pickle which successfully produced the same accuracies as calculated in question 3 of this section demonstrating reproducibility in the code. The accuracies can be seen below.

```
Random Forest Accuracy: 0.32976190476190476
XGBoost Accuracy: 0.4087301587301587
Ensemble Model Accuracy: 0.4158730158730159
```